COLLEGE TRANSPORT MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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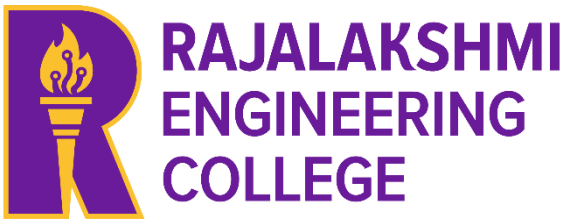
in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "COLLEGE TRANSPORT MANAGEMENT SYSTEM" is the bonafide work of LIVYA REMI ROSI E (240701289), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027 and Semester V, under my supervision.

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DR J. MANORANJINI HEAD OF THE DEPARTMENT DEPARTMENT OF CSE Rajalakshmi Engineering College Thandalam – 602 105	ASWANA LAL COURSE FACULTY ASSISTANT PROFESSOR DEPARTMENT OF CSE Rajalakshmi Engineering College

Submitted for the End Semester Examination held on [11/11/2026].

INTERNAL EXAMINER EXTERNAL EXAMINER

DECLARATION

I, LIVYA REMI ROSI E (240701289), hereby declare that this project report titled "COLLEGE TRANSPORT MANAGEMENT SYSTEM", submitted for the End Semester Examination of CB23F35 – Low-Code Application Development using Zoho Creator, is a record of project work carried out by me. The application described in this report is developed using Zoho Creator with forms, reports, pages and workflows wherever implemented.

The final report reflects the features identified in the available application evidence. AI-assisted configuration, sample data, code snippets and external resources used are acknowledged appropriately.

Place: Chennai

Date: _____

SIGNATURE OF THE STUDENT

LIVYA REMI ROSI E

240701289

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman, Thiru S. Meganathan; Chairperson, Dr. Thangam Meganathan; and Vice-Chairman, Mr. Abhay Shankar Meganathan, for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S. N. Murugesan, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. J. Manoranjini, Head of the Department, Department of Computer Science and Engineering (CSE), for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Aswana Lal, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation / Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources. I thank my family and friends for their unwavering support.

LIVYA REMI ROSI E

240701289

ABSTRACT

Educational institutions manage student transport registration, bus and driver information, route allocation, attendance, fee collection, complaints and incident reporting. When these activities are maintained separately through paper records or spreadsheets, administrators may face difficulties locating information, tracking registrations and maintaining consistent records. The College Transport Management System is designed as a centralised application using Zoho Creator to organise these activities.

The application contains twelve forms: Student Details, Bus Details, Driver Details, Bus Route Assignment, Transport Registration, Transport Fee Management, Bus Attendance, Complaint and Feedback, Emergency and Incident Report, Bus Document Alerts, Driver Document Alerts and Route Capacity Report. Lookup relationships connect related records where configured. Reports provide access to submitted records, while dashboard components are intended to support operational monitoring. The workflow overview observed during preparation showed two existing workflows: High Priority Incident Email Alert and a Transport Registration workflow. Additional document-expiry and fee automations are not treated as implemented until created and tested.

A Bus Details record was submitted successfully and was visible in its report. The High Priority Incident Email Alert workflow was shown as enabled with one action. However, the initial incident email test did not result in a received email. Therefore, email delivery and other pending features require further testing before final submission. The system aims to reduce fragmented record keeping and provide a single location for college transport administration.

PROJECT AT A GLANCE

Attribute	Details
Project Title	College Transport Management System
Domain	Education / College Transportation
Student	Livya Remi Rosi E
Register Number	[240701289]
Zoho Creator Edition	[Standard Edition]
Application Name	College Transport Management System
Live URL	[https://creatorapp.zoho.in/livyaremirosi.e.2024.cse1248/collegetransport-management-system/#Section_1]
Number of Forms	12
Reports	[23]
Pages / Dashboards	[1]
Workflows visible	High Priority Incident Email Alert; Transport Registration workflow
Integrations	Zoho Creator email notification configuration; delivery test pending
Roles / Profiles	[Not configured]
Deployment Date	[01/11/2026]

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DBMS	Database Management System
ER	Entity–Relationship
FR	Functional Requirement
HTML	HyperText Markup Language
ID	Identification
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
UI	User Interface
UAT	User Acceptance Testing
URL	Uniform Resource Locator
UX	User Experience

CHAPTER 1 INTRODUCTION

1.1 Background

Transportation is an important support service in colleges, particularly for students commuting from different locations. Transport administration maintains student information, bus details, driver details, route assignments, registrations, fee payments, attendance and incident records. These activities involve multiple related data items and require timely updates.

The College Transport Management System is designed to centralise these activities in a Zoho Creator application. The platform provides form-based data entry, lookup relationships, reports, pages and workflow automation. The application is intended for authorised transport administrators and staff.

1.2 Problem Statement

When transport information is maintained in separate registers or spreadsheets, it may be difficult to identify the latest student registration, bus allocation, route assignment, payment status or attendance record. Manual tracking can delay the handling of complaints, emergency incidents and document-expiry information. A centralised system is required to organise transport records and support timely administrative actions.

- Student, bus, driver and route information may be stored separately.
- Registration status and bus allocation need consistent tracking.
- Fee balances require accurate updates.
- Attendance, complaints and incidents need central recording.
- Bus and driver document validity needs monitoring.

1.3 Objectives

- Design a centralised data model for students, buses, drivers and routes.
- Provide forms for registration, fees, attendance and incident reporting.
- Connect related records using lookup fields and reports.
- Automate selected operations using workflows and Deluge.
- Provide operational information through reports and dashboards.
- Test form submission, record visibility and notifications.

1.4 Scope of the Project

The scope includes student and fleet data capture, route assignment, transport registration, fee records, bus attendance, complaints, emergency incidents and document-alert records. External GPS, payment gateway, SMS and third-party integrations are excluded unless separately configured and tested.

1.5 Justification for a Low-Code Approach

Zoho Creator provides visual form building, lookup relationships, reports, pages, workflows and Deluge scripting in one platform. These capabilities suit an administrative application centred on data-entry forms and workflow-driven processes. Low-code development reduces conventional coding effort and supports iterative changes.

1.6 Organisation of the Report

Chapter 2 studies the existing process and proposed solution. Chapter 3 presents requirements and user roles. Chapter 4 explains architecture, data model and workflow design. Chapter 5 documents implementation.

CHAPTER 2 LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional college transport process, students submit transport requests and staff collect student and route details. Staff maintain bus and driver information, allocate routes, record payments and track attendance. Complaints and incidents may be recorded separately and require followup.

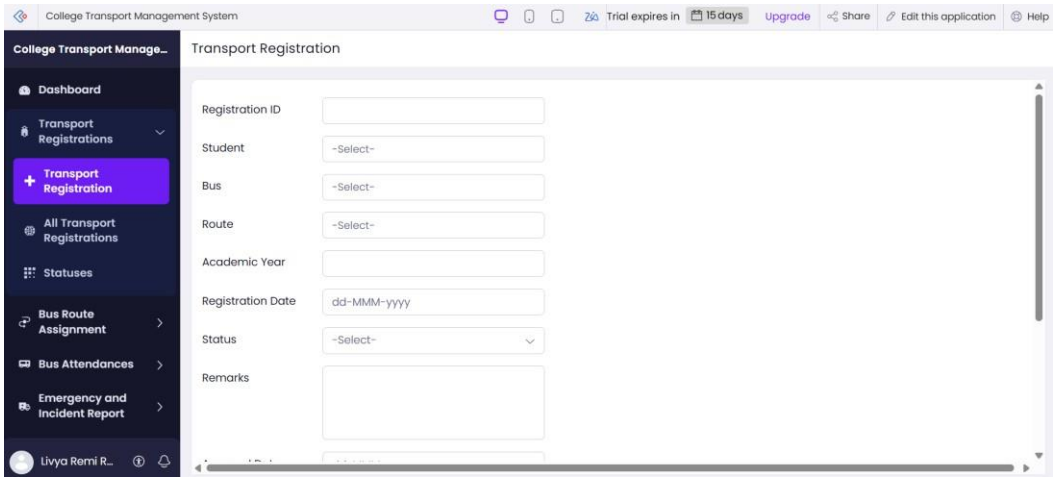


Figure 2.1: Existing College Transport Process (conceptual)

2.2 Limitations of the Existing System

- Information may be duplicated across files.
- Finding a student's current bus and route can require manual crosschecking.
- Fee balances and registration statuses may not update immediately.
- Attendance and incident information may be difficult to analyse centrally.
- Document-expiry follow-up may depend on manual reminders.

2.3 Low-Code versus Traditional Development

Criterion	Traditional Development	Low-Code on Zoho Creator
Development	Requires application code and database/UI implementation.	Uses visual builders and Deluge where needed.
Data capture	Forms must be designed and coded.	Forms are configured in the platform.
Reports	Custom report development required.	Reports and charts can be configured.
Automation	Backend logic and scheduling infrastructure required.	Workflows and Deluge support automation.
Maintenance	Changes may require code changes.	Many form/report changes use the builder.

2.4 Literature Survey

The project uses Zoho Creator's form, report, page and workflow capabilities. The final submission should cite at least five verified sources. The following are sources to consult, not claims that every item has already been reviewed.

S.No.	Source to consult	Relevance
1	Zoho Creator documentation – Forms	Form creation and field configuration.
2	Zoho Creator documentation – Reports	Report and chart configuration.
3	Zoho Creator documentation – Workflows	Trigger and action configuration.
4	Deluge documentation	Business rules and scripting.
5	Verified IEEE/ACM/Gartner low-code research	Academic background and comparison.

2.5 Emerging Trends in Low-Code Development

AI-assisted application building, visual development, workflow automation and reusable integrations are important low-code trends. Ziaassisted workflow creation may support configuration, but all generated workflows must be reviewed and tested.

2.6 Proposed System

The proposed system is a centralised Zoho Creator application containing twelve forms for transport administration. Lookup fields connect related records. Reports and dashboard components are intended to provide visibility into registrations, fleet information, attendance, payments and incidents.

CHAPTER 3 REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities

Administrator	Maintains application and transport records.	Manage forms, reports, settings and workflows.
Transport Staff	Handles day-to-day operations.	Enter student, bus, driver, route, attendance and fee records.
Student / Applicant	Transport service user if portal access is configured.	Submit or view transport information only if access is enabled.

Exact roles and permission profiles configured in the live application must be confirmed before final submission.

3.2 Functional Requirements

ID	Requirement	Priority
FR-01	Store student details in a structured form.	High
FR-02	Maintain bus and driver information.	High
FR-03	Support route assignment and transport registration.	High
FR-04	Maintain transport fee and payment records.	High
FR-05	Record bus attendance.	High
FR-06	Capture complaints and emergency incidents.	High
FR-07	Maintain bus and driver document-alert records.	Medium
FR-08	Provide reports for stored transport data.	High
FR-09	Send configured email notifications for qualifying incidents.	High

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Forms and reports should be understandable to authorised staff.
NFR-02	Security	Access should be limited according to configured permissions.
NFR-03	Availability	Application depends on Zoho Creator service and internet connectivity.
NFR-04	Maintainability	Forms, reports and workflows should be maintainable in Creator.
NFR-05	Data integrity	Required fields and lookups should support consistent records.
NFR-06	Auditability	Record changes should be traceable where audit features are enabled.

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (edition to be confirmed)
Scripting	Deluge, where configured
Browser	Google Chrome / Microsoft Edge
Connectivity	Internet connection
Device	Desktop or laptop
Other services	Zoho Creator email notification service; delivery test pending

3.5 Use Case Model

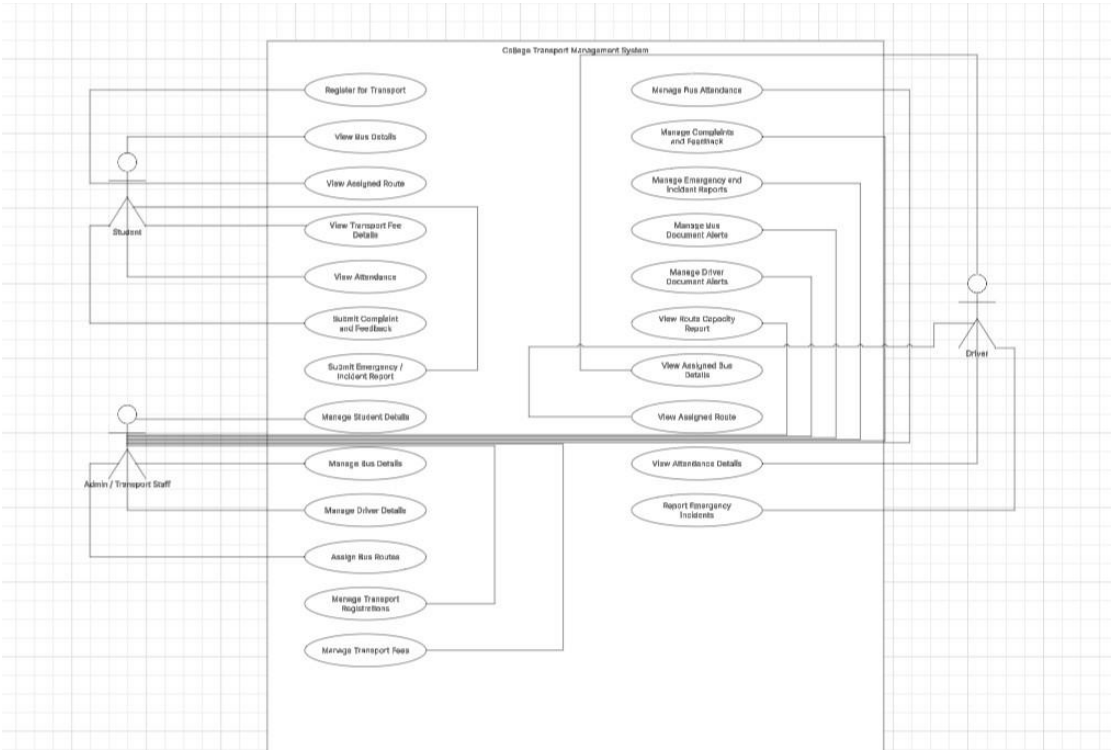


Figure 3.1: Use Case Diagram

Use Case Item	**Description**
ID / Name	UC-01 – Manage Transport Registration
Actors	Transport Administrator / Staff
Pre-condition	Student and bus/route details are available.
Main Flow	Open form → Enter student and transport details → Submit → Review saved record.
Alternate Flow	Complete missing mandatory details before submission.
Post-condition	Transport registration record is stored.
Related FR	FR-03

CHAPTER 4 SYSTEM DESIGN

4.1 System Architecture

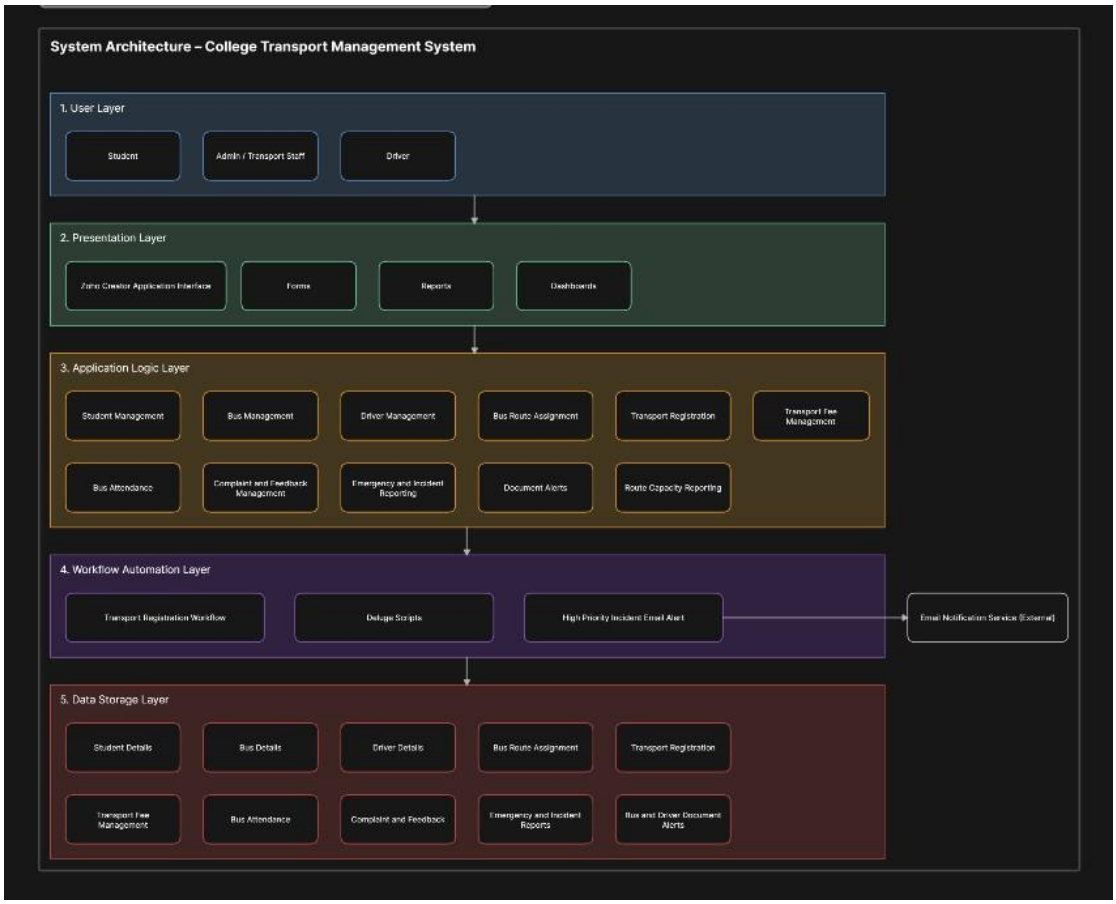


Figure 4.1: System Architecture

Users access the application through Zoho Creator. Forms capture transport data, reports display stored records and pages may combine reports and visual components. Workflows and Deluge scripts implement business rules and notifications where configured.

4.2 Data Model

The application contains twelve forms. Lookup fields connect related records, such as a student in a transport registration or a bus in a route assignment. Exact lookup relationships and field link names must be verified from the live form builder.

Form / Entity	Purpose
Student Details	Student master data.
Bus Details	Fleet and capacity information.
Driver Details	Driver information.
Bus Route Assignment	Bus, driver and route mapping.
Transport Registration	Student transport request/allocation.
Transport Fee Management	Fee and payment records.
Bus Attendance	Attendance entries.
Complaint and Feedback	Transport-related complaints.
Emergency and Incident Report	Incident details and severity.
Bus Document Alerts	Bus document-expiry information.
Driver Document Alerts	Driver document-expiry information.
Route Capacity Report	Route-capacity monitoring.

4.3 Form Design

Twelve forms are present. The following table summarises their intended purpose. Exact field labels, field types, mandatory settings, defaults and link names must be copied from the live form builder.

Form	Purpose	Key data to verify
Student Details	Student master	Name, roll number, department, year, contact
Bus Details	Fleet master	Bus number, registration details, seating capacity
Driver Details	Driver master	Driver name, phone, licence details
Bus Route Assignment	Bus/driver/route mapping	Bus, driver, route, assignment
Transport Registration	Student allocation	Student, bus, route, academic year, status
Transport Fee Management	Fee tracking	Total fee, paid amount, balance, status
Bus Attendance	Attendance	Student, bus, date, attendance status

Complaint and Feedback	Issue capture	Complaint details, category, status
Emergency and Incident Report	Incident capture	ID, type, severity, location, date/time
Bus Document Alerts	Bus document validity	Bus, document type, expiry date
Driver Document Alerts	Driver document validity	Driver, document type, expiry date
Route Capacity Report	Capacity monitoring	Route/bus capacity and registration count

4.4 Process and Workflow Design

The workflow list observed in the application showed two existing workflows: High Priority Incident Email Alert and a Transport Registration workflow. The incident alert was shown as enabled with one action. The exact trigger, condition and action for the Transport Registration workflow must be captured from its edit screen.

Workflow	**Trigger / Type**	**Action / Condition**	**Status**
High Priority Incident Email Alert	Created / successful form submission	High/Critical incident email alert intended.	Enabled; 1 action visible; delivery not confirmed.
Transport Registration workflow	Existing form workflow	Exact logic to be checked in editor.	Listed; verify enabled state.
Bus Document Expiry Reminder	Proposed scheduled workflow	30-day and expireddocument reminders.	Not counted as implemented.
Driver Document Expiry Reminder	Proposed scheduled workflow	30-day and expireddocument reminders.	Not counted as implemented.

4.5 User Interface and Navigation Design

Navigation is organised around the transport modules. Forms are used for data entry and reports for viewing records. The intended dashboard may present registrations, bus

capacity, route assignments, attendance, complaints and incidents. Insert only the actual live dashboard screenshot.

4.6 Access Control Design

The permission matrix below must be completed using the actual roles and access settings in Zoho Creator.

Component	Administrator	Transport Staff	Student / Portal
Student Details	[Available]	[Not Verified]	[N/C]
Bus / Driver Details	[Available]	[Not Verified]	[N/C]
Transport Registration	[Available]	[Not Verified]	[N/C]
Fee Management	[Available]	[Not Verified]	[N/C]
Attendance / Incidents	[Available]	[Not Verified]	[N/C]
Reports / Dashboard	[Available]	[Not Verified]	[N/C]

CHAPTER 5 IMPLEMENTATION

5.1 Application Setup

The College Transport Management System was created in Zoho Creator and contains forms for major transport administration functions. Add the application edition, theme, URL and deployment settings from the live application.

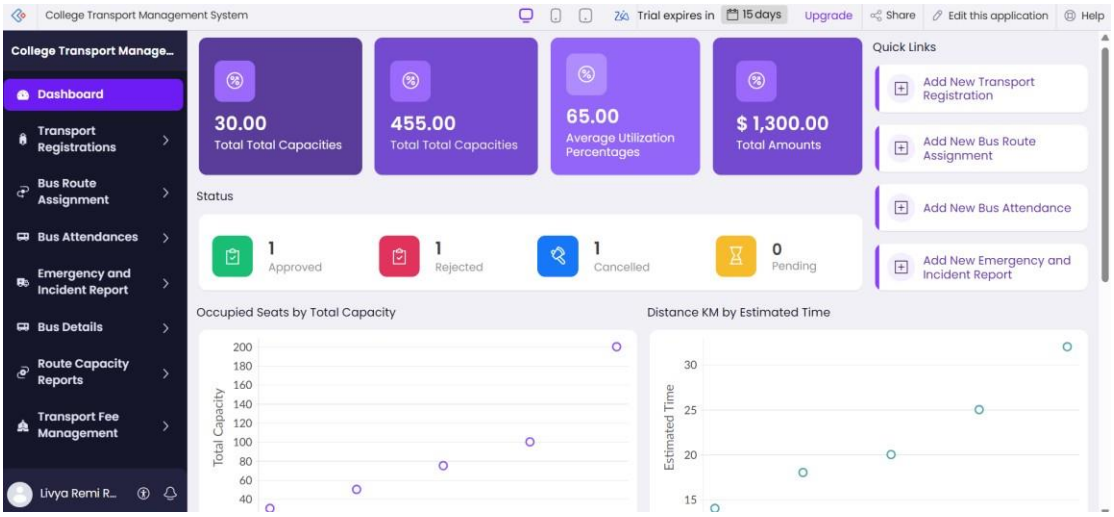


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Twelve forms are present: Student Details, Bus Details, Driver Details, Bus Route Assignment, Transport Registration, Transport Fee Management, Bus Attendance, Complaint and Feedback, Emergency and Incident Report, Bus Document Alerts, Driver Document Alerts and Route Capacity Report.

The Bus Details form was tested by submitting a bus record. The interface displayed a successful-save message and the record was subsequently visible in the Bus Details report.

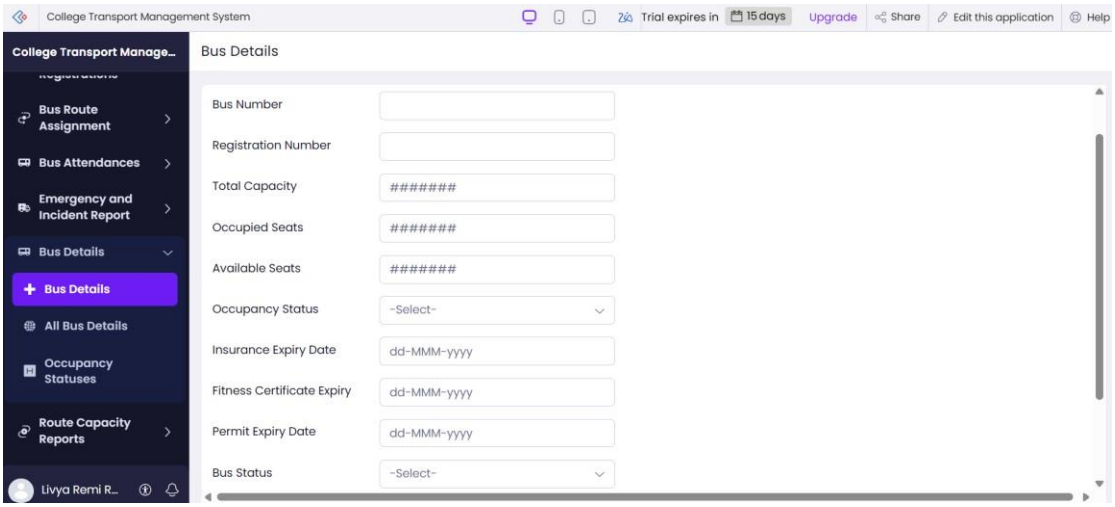


Figure 5.2: Bus Details Form and Saved Record

5.3 Reports

Reports provide access to records stored through the forms. The Bus Details report was used to confirm that a submitted bus record was visible. Other report names, types, filters and grouping options must be listed after checking the live application.

Report	Purpose	Verification
Bus Details report	View saved bus records.	Record visibility confirmed.
All Transport Registrations	View student transport registrations.	Verify columns and layout.
Transport Fee reports	View payment and balance records.	Verify actual names and output.
Other reports	Attendance, complaints, incidents and documents.	Add exact names from app.

5.4 Pages and Dashboards

The intended dashboard provides a consolidated view of transport operations. Suggested measures include total students, active buses, drivers, routes, seat capacity, occupied seats, available seats, pending registrations, attendance, complaints and incidents. Mark only components visible in the live dashboard as implemented.

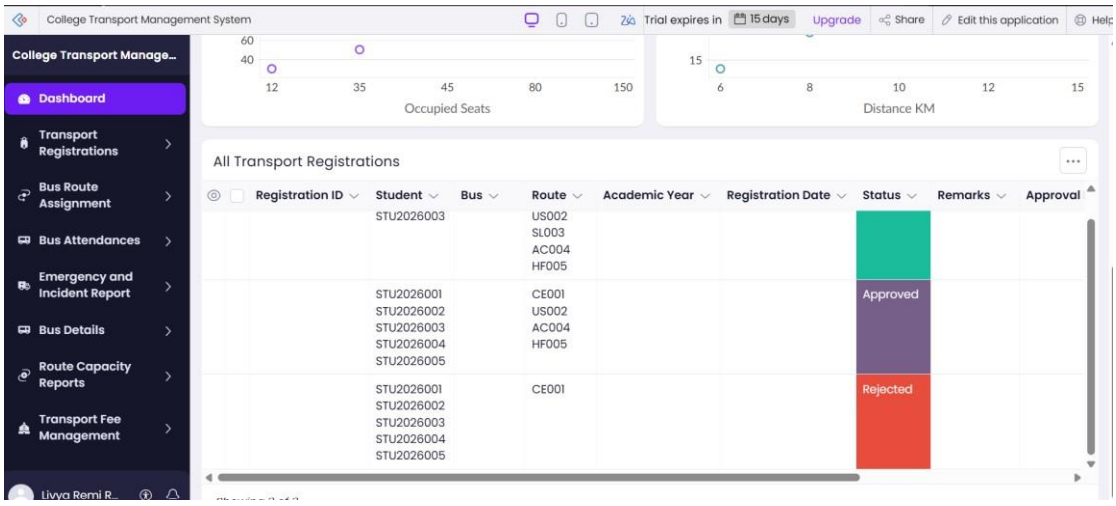


Figure 5.3: Transport Management Dashboard

5.5 Deluge Scripting

Deluge may be used for form validation, calculations and email notification actions. Complete source code and field link names must be copied from the live

workflow/script editor. Untested sample code must not be represented as implemented code.

Script / Action	Location	Purpose	Status
Incident email action	Emergency and Incident Report workflow	Alert for qualifying severity.	Workflow enabled; delivery not confirmed.
Registration action	Transport Registration workflow	Existing registration automation.	Inspect and document exact logic.
Fee balance calculation	Transport Fee Management	Calculate balance and payment status.	Proposed; not confirmed.
Document expiry reminder	Bus / Driver Document Alerts	Notify before/after expiry.	Proposed; not confirmed.

5.6 Workflows and Automation

At the time of preparing this draft, the workflow overview displayed two existing workflows. High Priority Incident Email Alert was listed with one action and an Enabled status. The Transport Registration form was also listed under forms with workflows. The exact trigger and action details should be recorded from the workflow edit pages.

The incident notification is intended to send an email to the configured administrator address when an incident is classified as High or Critical. A test incident record was saved successfully, but the expected email was not received during the initial test. Email delivery therefore remains pending and must not be reported as successful until verified.

Bus and driver document-expiry reminders are recommended because dedicated document-alert forms exist. They are not included in the implemented workflow count until created, enabled and tested.

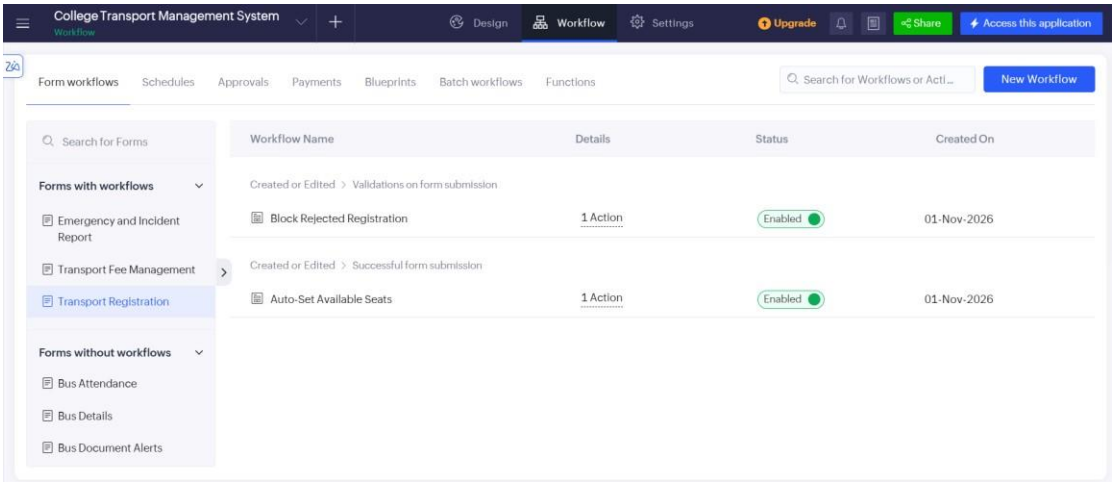


Figure 5.4: Workflow Configuration

5.7 Integrations

The application uses Zoho Creator email notification capability for incident alerting. The sender account was reported as verified and a recipient Gmail address was configured. However, the test notification was not received, so successful email integration has not yet been demonstrated.

5.8 User Management and Role-Based Access

Access should be restricted through Zoho Creator user and permission settings. Add screenshots and a role-permission matrix only after verifying configured roles and profiles.

5.9 AI and Analytics Features

Zia AI was used as an assistant to help create or configure workflow requirements. Generated workflows must be reviewed in the builder and tested. Analytics features should be documented only when actual charts, pivot reports or dashboard components are available.

5.10 Pro-Code Extension

No external pro-code extension is confirmed in this draft. The application primarily uses Zoho Creator's low-code form, report and workflow facilities.

5.11 Mobile and Portal Access

Mobile and portal access have not been confirmed. If the application is accessed through the Zoho Creator mobile application or a customer portal, insert screenshots and describe configured permissions.

CHAPTER 6 TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is performed by entering sample records, submitting forms, checking the success response and verifying that records appear in their corresponding reports. Workflow testing separately verifies trigger, condition, action and resulting notification.

6.2 Test Cases

Test ID	Scenario	Expected Result	Observed Result / Status
TC-01	Submit Bus Details record.	Record saved.	Success message and report visibility confirmed.
TC-02	Submit incident with High severity.	Record saved and email sent.	Record saved; email not received. Pending.
TC-03	Submit Critical incident.	Email alert sent.	Not yet confirmed.
TC-04	Submit transport registration.	Record stored and workflow runs.	Verify live workflow and report.
TC-05	Update transport fee record.	Balance/status correct if automation exists.	Not yet confirmed.
TC-06	Check document nearing expiry.	Reminder sent if schedule exists.	Not yet implemented/verified.
TC-07	Open reports for saved records.	Records visible.	Bus Details confirmed; others to verify.

6.3 Debugging Deluge Scripts

The incident email action requires verification of sender, recipient, trigger, severity condition and email task execution. A successful form submission confirms that a record was added; it does not prove that the email action ran. Workflow execution history and email configuration should be checked.

6.4 Defect Log

Defect	Issue	Status	Action Required
D-01	Incident record saved but email not received.	Open	Check workflow execution, sender, recipient and email task.
D-02	Registration report fields previously appeared blank or multiplevalued.	Verify	Inspect lookup fields, link names and report columns.
D-03	Workflow overview shows only two workflows.	Current state	Create/test further workflows only after checking fields.

6.5 User Acceptance Testing

User acceptance testing should include representative student, bus, registration, attendance, fee and incident records. Confirm that records are stored, displayed in the intended report and accessible only to authorised users.

CHAPTER 7 DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is being configured in Zoho Creator. Production deployment and published URL have not been confirmed. Before deployment, verify forms, reports, workflows, permissions and test records.

Checklist	Status
Forms and fields verified	[Verified]
Reports and dashboard verified	[Verified]
Workflows enabled and tested	[Verified]
Email sender and recipient	Sender reported verified; delivery pending
User permissions checked	[Verified]
Published URL	[Published]

7.2 Performance Optimisation

Use appropriate lookup fields, report filters and concise dashboard components. Large reports should be filtered by date, route, bus or status where suitable. Performance measurements must be collected from actual use.

7.3 Security Features

- Use verified user accounts and strong authentication.
- Restrict student, driver and payment information according to role.
- Avoid public sharing of sensitive reports.
- Use authorised recipients for incident notifications.
- Review permissions and audit information periodically.

7.4 Governance and Administration

The administrator should maintain form structure, report definitions, workflow rules and user access settings. Changes should be tested before use with operational records. Backup and retention should follow institutional requirements and the selected Creator plan.

CHAPTER 8 RESULTS AND DISCUSSION

8.1 Application Output

The application provides twelve transport-related forms. A Bus Details record was submitted successfully and was visible in its report. The workflow overview showed two existing workflows, including an enabled High Priority Incident Email Alert. The initial incident email test did not result in a received email, so notification delivery remains unresolved.

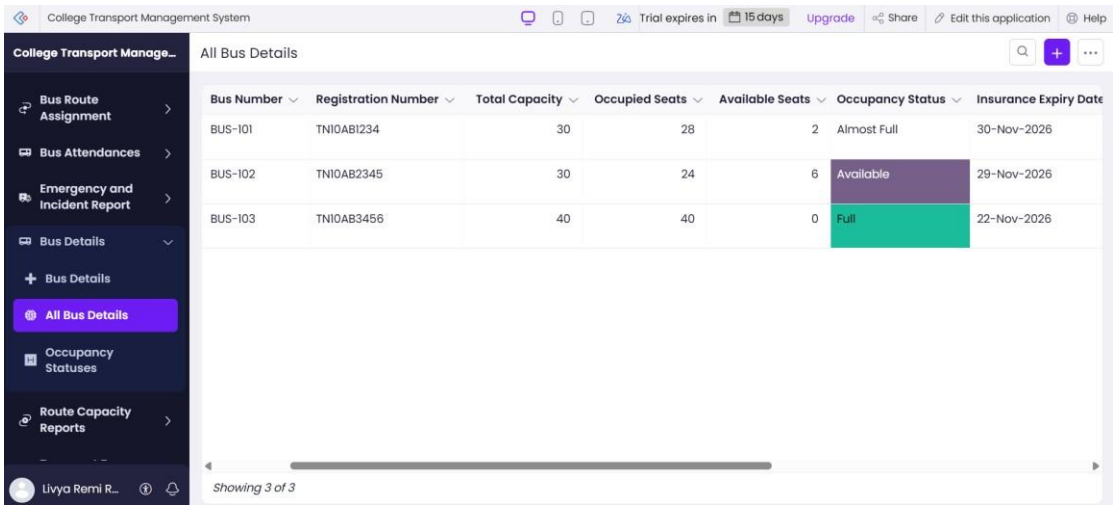


Figure 8.1 a) Bus Details Outputs

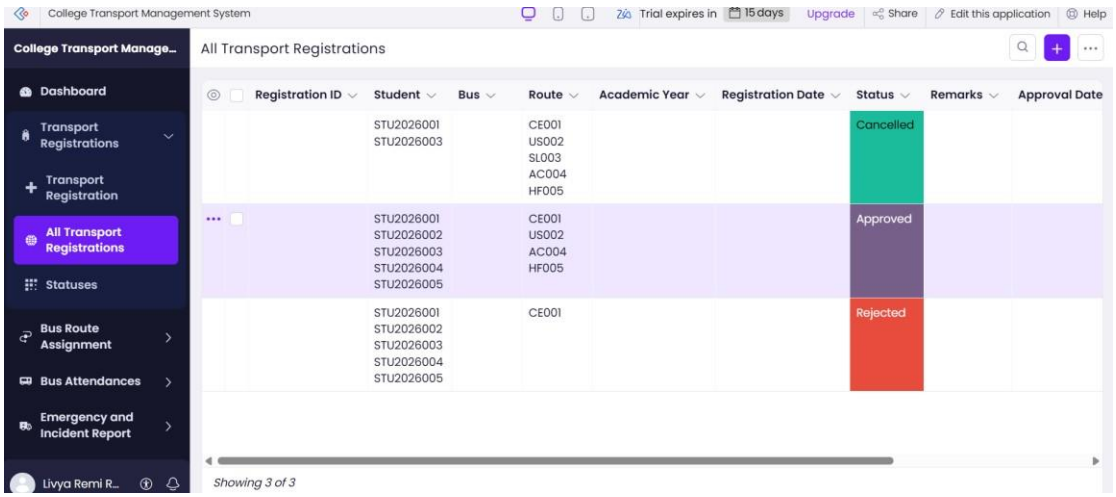


Figure 8.1 b) Transport Registration Outputs

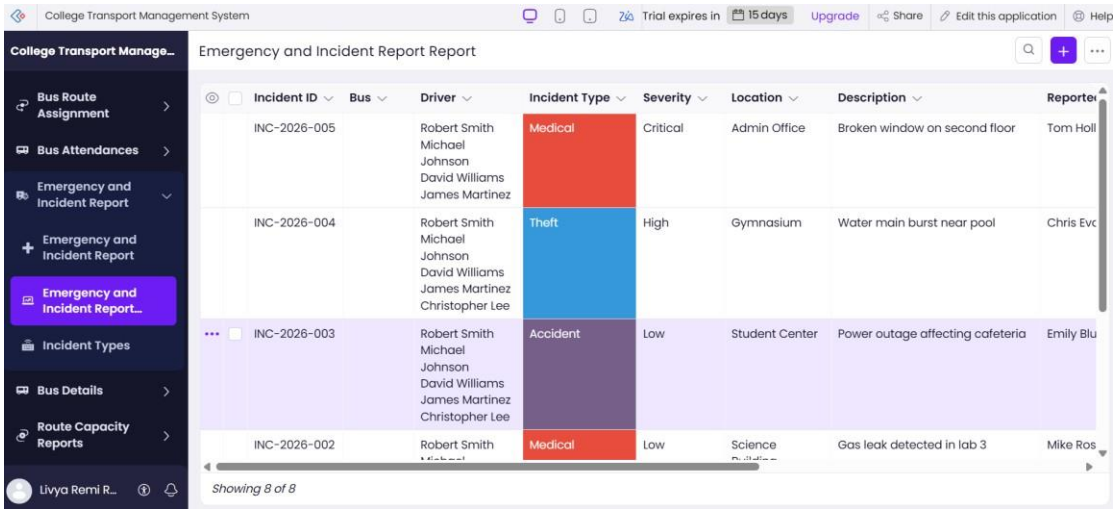


Figure 8.1 c) Emergency and Incident Report Outputs

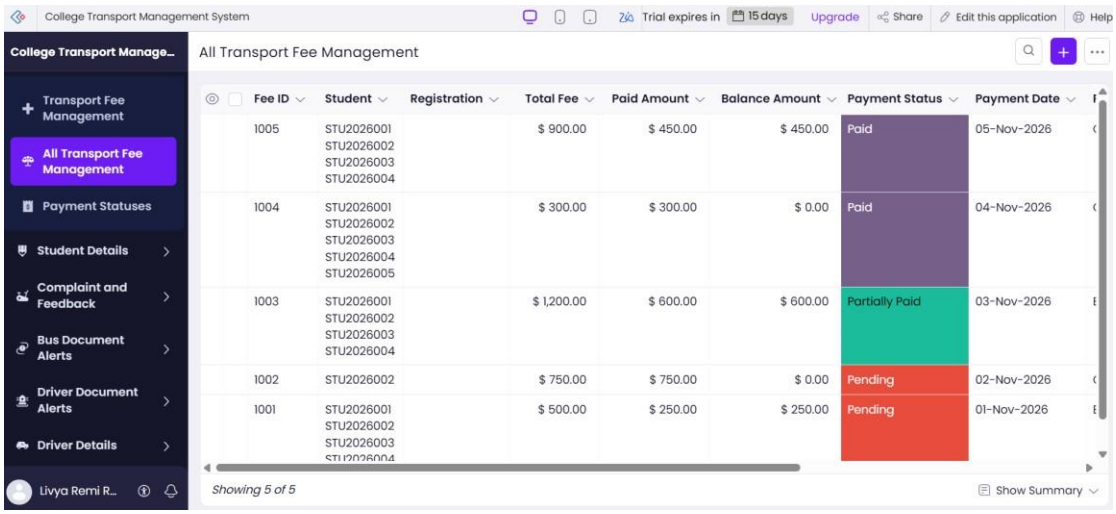


Figure 8.1 d) Transport Fee Management Outputs

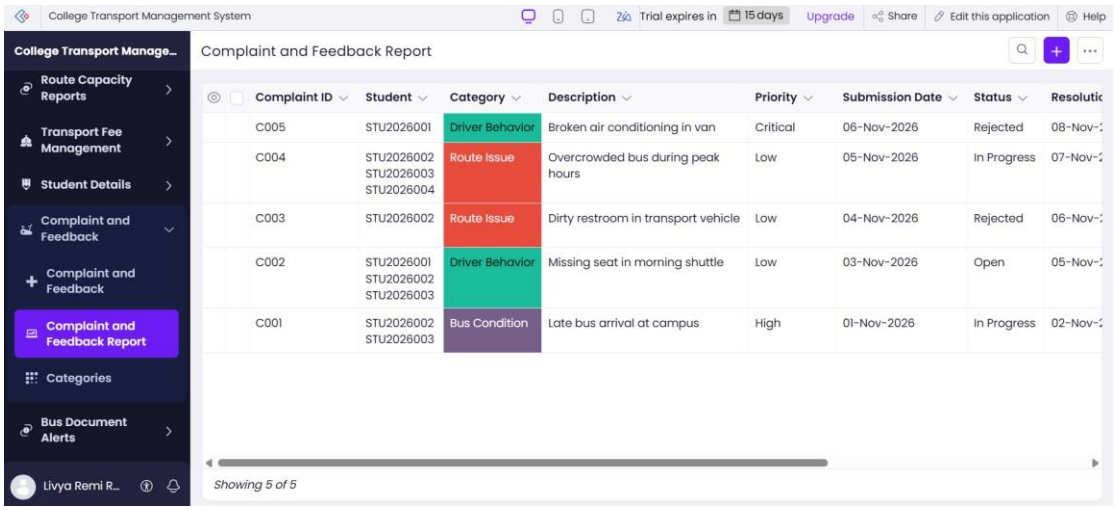


Figure 8.1 e) Complaint and Feedback Reports Outputs

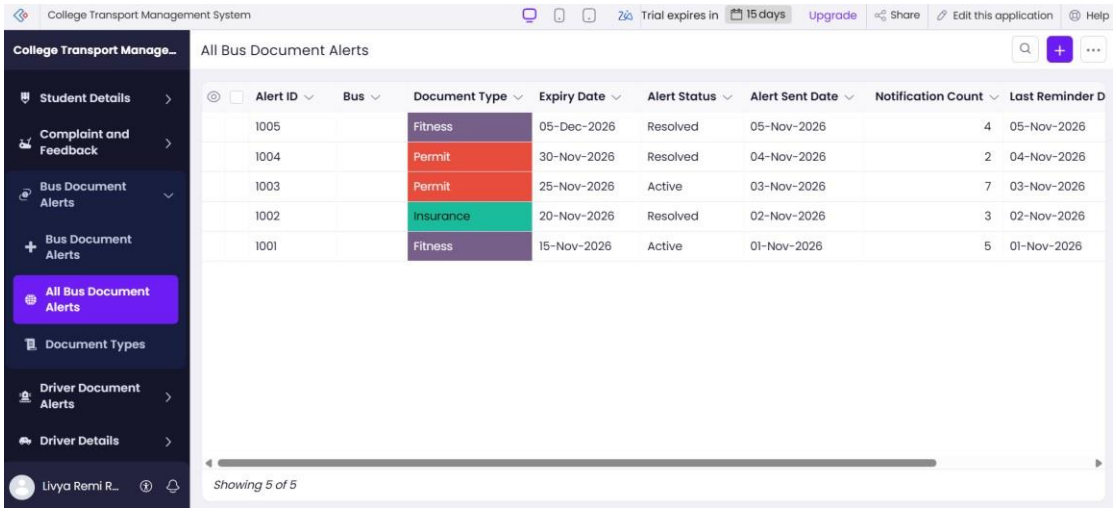


Figure 8.1 f) Bus Documents Outputs

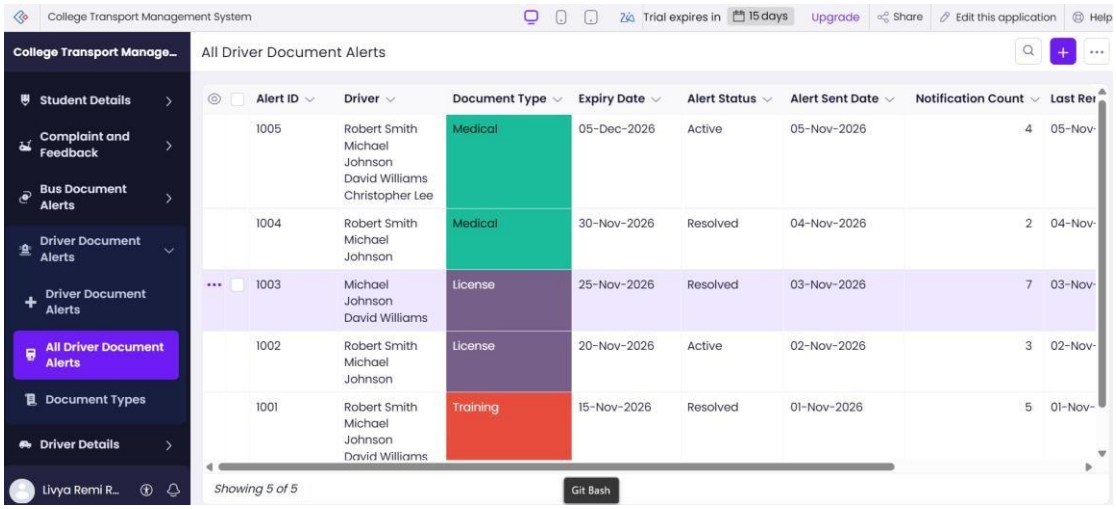


Figure 8.1 g) Driver Document Outputs

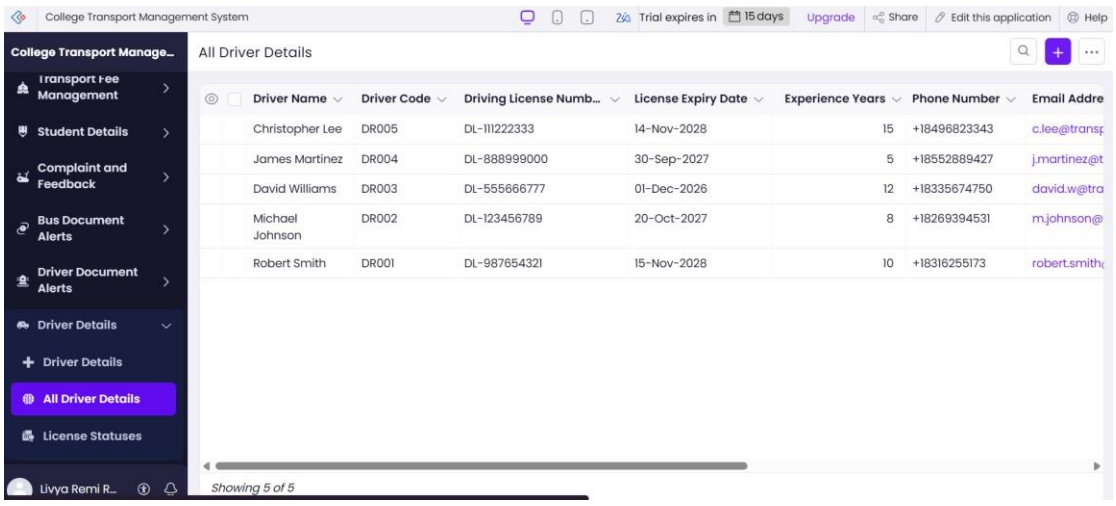


Figure 8.1 h) Driver Details Outputs

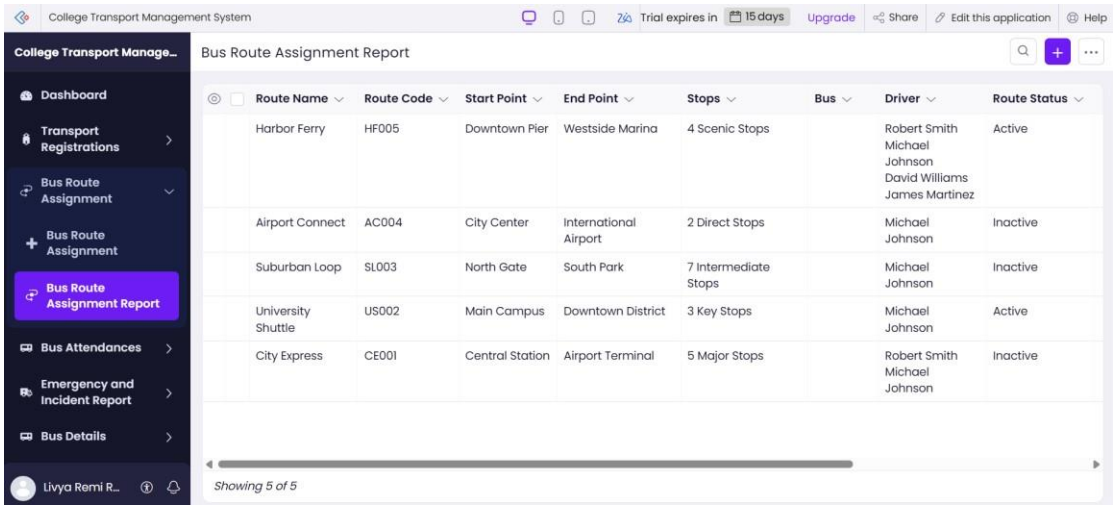


Figure 8.1 i) Bus Route Assignment Report Outputs

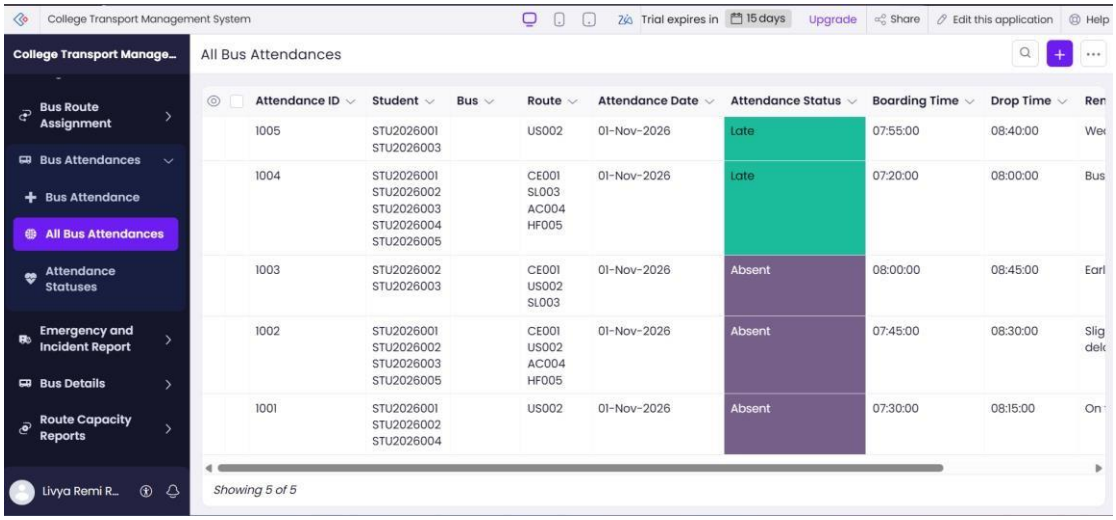


Figure 8.1 j)Bus Attendance Outputs

Figure 8.1: Verified Application Outputs

8.2 Achievement of Objectives

Objective	Evidence	Status
Create centralised transport forms	Twelve forms listed.	Form presence confirmed; field review pending.
Maintain bus records	Bus record saved and visible.	Verified.
Support transport registration	Form and workflow listed.	Further testing required.

Automate incident notification	Enabled workflow visible.	Email delivery pending.
Monitor fees, attendance and documents	Dedicated forms exist.	Automation/output testing pending.

8.3 Comparison with Existing System

Aspect	**Manual / Separate Records**	**Creator Application**
Data storage	Separate files/registers.	Twelve forms in one app.
Record retrieval	Manual searching.	Reports display submitted records.
Notifications	Manual follow-up may be needed.	Configured workflows can trigger notifications.
Operational visibility	Manual consolidation.	Reports/dashboard can consolidate data when configured.

8.4 Limitations of the Low-Code Approach

The application depends on the selected Zoho Creator edition and platform availability. Workflow execution and email delivery depend on correct configuration and verified sender details. Advanced integrations may require additional Deluge scripting or external services.

8.5 Reflection on No-Code, Low-Code and Pro-Code

The application uses visual configuration for forms and reports, workflow configuration for automation and Deluge scripting where custom logic is required. No external pro-code extension is confirmed.

CHAPTER 9 CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The College Transport Management System provides a centralised Zoho Creator application structure for student, bus, driver, route, registration, fee, attendance, complaint, incident and document-alert information. Twelve forms are present, and the successful saving and retrieval of a bus record demonstrate basic data-capture capability. The workflow overview shows existing automation for incident alerting and transport registration. Further testing is required to confirm email delivery and the complete behaviour of each workflow.

9.2 Future Enhancements

- Complete and test bus and driver document-expiry reminder workflows.
- Verify transport registration approval and capacity validation.
- Implement and test fee balance and payment-status automation if required.
- Add route-wise and bus-wise occupancy analytics.
- Provide role-based access for administrators, staff and students where appropriate.
- Resolve and test incident email delivery.
- Consider GPS tracking or payment gateway integration only if required and supported.

9.3 Learning Outcomes and Career Relevance

This project provides practical experience in low-code application development, data modelling, lookup relationships, report creation, workflow configuration, Deluge automation, debugging and testing.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help – Forms,” official product documentation. [Insert verified URL and access date].

[2] Zoho Corporation, “Zoho Creator Help – Reports,” official product documentation. [Insert verified URL and access date].

[3] Zoho Corporation, “Zoho Creator Help – Workflows,” official product documentation. [Insert verified URL and access date].

[4] Zoho Corporation, “Deluge Help,” official scripting documentation. [Insert verified URL and access date].

[5] [Add a verified IEEE/ACM/Gartner research source on low-code application platforms].

[6] [Add a verified source related to transport or fleet management systems].

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Feature	Use in College Transport Management System	Status
Application and Navigation	Forms, reports and dashboard for transport management	Used – Chapter 5
Forms and Field Types	Student, Bus, Driver and Transport-related forms	Used – Chapter 5
Lookup Relationships	Connecting student, bus, driver and route records	Pending verification
Validation Rules	Validating transport registration and form inputs	Pending verification
List Reports	Viewing bus, student and transport records	Used – Section 5.3
Charts and Dashboard	Displaying transport and payment information	Partially implemented
Deluge Form Workflows	Automating form-related operations	Used – Appendix C
Deluge Record Fetch / Update	Fetching and updating related transport records	Pending verification
Notifications	High-priority incident email alert	Configured – delivery to be verified
Approval Process	Transport registration workflow	Configured – verification required
Subform and Advanced Reports	Subforms, Kanban and Calendar reports	Not confirmed
Scheduled and Batch Workflows	Scheduled or batch-based automation	Not confirmed
External Integration and Zia / AI	External services and AI-based features	Not confirmed
Access Permissions	Role-based access to forms and reports	Pending verification
Production Deployment, Portal and Mobile	Production, portal and mobile access	Not confirmed

APPENDIX B – COURSE OUTCOME MAPPING

B.1 Course Outcome Mapping Table

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture and business value of low-code development using Zoho Creator.	Chapters 1, 2 and 4.1 – Introduction, system study and architecture.	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows and Deluge scripting.	Chapters 4, 5.1–5.6 and 6 – Design, implementation, workflows and testing.	Good	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections and RBAC.	Chapters 5.7, 5.8 and 7.3. Live integration and access-control implementation require verification.	Basic – pending verification	
CO4	Deploy, secure and manage Zoho Creator applications across development and production environments following governance best practices.	Chapter 7 – Deployment, Security and Governance. Production deployment and environment configuration are not confirmed.	Basic – pending verification	
CO5	Evaluate low-code development strategies, AI-assisted application development and the transition from citizen developer.	Chapters 2.3, 8.4, 8.5 and 9.3 – comparison, limitations and learning outcomes.	Good	

B.2 Course Outcome Achievement Summary

CO	Achievement Description	Project Evidence
CO1	Understanding of low-code development concepts and Zoho Creator architecture.	Design and explanation of the College Transport Management System.
CO2	Development of application forms, reports and workflows.	Twelve application forms and the workflows visible in the application.
CO3	Understanding of integration and access-control concepts.	Integration and access-control design discussion; live implementation requires verification.
CO4	Understanding of deployment, security and governance requirements.	Deployment and security discussion; production deployment is not confirmed.
CO5	Evaluation of low-code development benefits, limitations and future possibilities.	Project evaluation, limitations and future enhancement discussion.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 High Priority Emergency Incident Email Alert

The following is a previously discussed draft for the High Priority Incident Email Alert workflow. It is included as a source-code reference, not as a verified export of the live script. Verify all form link names, field link names, lookup field data types and recipient details in Zoho Creator before use.

Purpose: Check incident severity and prepare an email notification when the severity is High or Critical.

```
// Draft Deluge script – verify link names against the actual application

incidentRecord = emergency_incident[ID == input.ID];

if(incidentRecord.severity == "High" || incidentRecord.severity == "Critical")
{
emailMessage = "<h2>High Priority Emergency Incident Alert</h2>";
emailMessage = emailMessage + "<p><strong>Incident ID:</strong> " +
incidentRecord.incident_id + "</p>";
emailMessage = emailMessage + "<p><strong>Incident Type:</strong> " +
incidentRecord.incident_type + "</p>";
emailMessage = emailMessage + "<p><strong>Severity:</strong> " +
incidentRecord.severity + "</p>";
emailMessage = emailMessage + "<p><strong>Location:</strong> " +
incidentRecord.incident_location + "</p>";
emailMessage = emailMessage + "<p><strong>Description:</strong> " +
incidentRecord.description + "</p>";
emailMessage = emailMessage + "<p><strong>Reported By:</strong> " +
incidentRecord.reported_by + "</p>";
emailMessage = emailMessage + "<p><strong>Date & Time:</strong> " +
incidentRecord.incident_datetime + "</p>";

if(!isNull(incidentRecord.bus))
{
busRecord = bus_details[ID == incidentRecord.bus];
emailMessage = emailMessage + "<p><strong>Bus Number:</strong> " +
busRecord.bus_number + "</p>";
}

if(!isNull(incidentRecord.driver))
{
driverRecord = driver_details[ID == incidentRecord.driver];
emailMessage = emailMessage + "<p><strong>Driver Phone:</strong> " +
driverRecord.phone_number + "</p>";
}
```

```
sendmail
[
from : zoho.adminuserid
to : "[Verified recipient email]"
subject : "URGENT: High Priority Emergency Incident"
message : emailMessage
content type : HTML
];
}
```

C.2 Script Configuration Details

Item	Configuration / Verification
Workflow name	High Priority Incident Email Alert.
Trigger	Verify the configured form event and execution condition in Zoho Creator.
Severity condition	High or Critical.
Sender	zoho.adminuserid; sender verification and account configuration must be checked.
Recipient	Enter the verified recipient email configured for the project.
Email delivery status	Pending confirmation; do not claim successful delivery until a test email is received.
Form and field link names	Verify emergency_incident, severity, incident_id, incident_type, incident_location, description, reported_by, incident_datetime, bus, driver, bus_details, bus_number, driver_details and phone_number against the actual application.

C.3 Important Verification Note

This appendix contains the previously discussed draft script, not a verified export of the live Zoho Creator workflow. The script may require changes to match the application's exact Deluge form and field link names and lookup-field data types. Test with both High and Critical severity records, and confirm the workflow execution log and receipt of the email before marking the notification feature as fully implemented.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

The College Transport Management System is developed using Zoho Creator. An authorised user can access the application through the application link shared by its owner or administrator. The user must sign in with the permitted Zoho account, if authentication is enabled.

Steps to access the application:

- Open a web browser on a computer or mobile device.
- Enter the College Transport Management System application URL provided by the application owner.
- Sign in using the authorised Zoho account, if prompted.
- Open the application home page and select the required form or report from the navigation menu.
- Use only the modules and records permitted by the access settings configured for the account.

Note: The live application URL and deployment status must be added after the application is published and the URL is confirmed.

D.2 Instructions for Role 1 – Application Administrator / Creator

This section is intended for the person who owns or administers the Zoho Creator application. The actions below are available only when the signedin account has the required permissions.

Step	Module / Task	User Instructions
1	Open the application	Sign in to Zoho Creator and open the College Transport Management System.
2	Manage student records	Open Student Details. Add a student record, enter the required information, review the entries and save.
3	Manage bus records	Open Bus Details. Enter bus information and save. Confirm the record appears in the relevant report.
4	Manage driver records	Open Driver Details and enter driver information using the fields provided.
5	Assign routes	Open Bus Route Assignment and enter route allocation details. Check the selected bus and related information before saving.
6	Review transport registrations	Open Transport Registration to view or record student transport registration details.
7	Manage fee information	Open Transport Fee Management to enter or review transport fee records.
8	Record attendance	Open Bus Attendance and enter attendance information for the relevant journey or date.
9	Review complaints	Open Complaint and Feedback to record and review submitted complaints or feedback.
10	Record incidents	Open Emergency and Incident Report to enter incident details and select the appropriate severity.
11	Review document alerts	Use Bus Document Alerts and Driver Document Alerts to maintain or review document-related alert records.
12	Review route capacity	Open Route Capacity Report to view or maintain route capacity information as supported by the configured form.
13	Check workflows and reports	Review available reports and workflow configuration. Confirm workflow execution through available history or logs.

D.3 Instructions for Role 2 – Authorised Application User

This section is for a staff member or other authorised user who has been granted access to the application. The project currently does not confirm separately configured Student, Driver, or Transport Staff profiles. Therefore, the exact actions available to this user depend on the permissions actually assigned by the application owner.

Step	Task	Instructions
1	Sign in	Access the application using the URL and Zoho account authorised by the application owner.
2	Open permitted modules	Use the navigation menu to open only forms and reports available to your account.
3	Enter information	Complete required fields in the relevant form. Check spelling, dates and selected values before saving.
4	Save and verify	Submit the form and look for the save confirmation. Where permitted, open the corresponding report to confirm the record is listed.
5	Report an issue	Use Complaint and Feedback or Emergency and Incident Report only if those forms are available and you are authorised.
6	Protect information	Do not access, edit, download or share records unless permitted. Sign out after completing your work.

D.4 General Guidelines

- Enter accurate and complete information in all required fields. Avoid creating duplicate student, bus, driver, registration, or fee records.
- Verify the selected bus, route, and driver details before saving related records.
- Use the correct incident severity when entering an emergency or incident report.
- Do not assume that an email alert has been delivered unless receipt is confirmed.
- Contact the application owner if a form, report, or action is not visible or accessible.

APPENDIX E – EVALUATION RUBRIC

Student Name	Livya Remi Rosi E	Register No.	240701289
Project Title	College Transport Management System	Date	11/11/2026

Table E.1: Evaluation Rubric for the Individual Project

Criterion	**Max Marks**	**Awarded**
Problem definition and requirements	10	
System design	10	
Forms, reports and pages	15	
Deluge and workflows	15	
Integration, RBAC and AI	10	
Testing, deployment and governance	10	
Live demonstration	15	
Report	5	
Viva voce	10	
TOTAL	**100**	

APPENDIX F – VIVA-VOCE RECORD SHEET

Student:LIVYA REMI ROSI E Date:11/11/2026

Register Number: 240701289

Table F.1: Viva-Voce Record

Question	Response / Remarks
1	
2	
3	
4	
5	
6	
7	
8	

INTERNAL EXAMINER EXTERNAL EXAMINER